Arista Access Node (MALL-ACC-21) Optical Transport Failure SOP

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7613

Description

The failure of Optical Transport on the Arista Access Node (MALL-ACC-21) in the Malleswaram Node can be attributed to the degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This SOP outlines the steps necessary to diagnose, resolve, and verify the fix for this failure domain. In an Arista network, the Access Node (MALL-ACC-21) is responsible for connecting customer-facing devices to the core network. The Optical Transport Failure can occur due to issues with the SFP/XFP modules, which are responsible for transmitting and receiving optical signals. The Rx/Tx power loss can be attributed to faulty or damaged fiber optic cables or connectors. The fiber cuts can be caused by physical damage to the cables or connectors. The Arista CLI provides a set of commands to diagnose and resolve these issues, including the 'show interface' command to check the status of the SFP/XFP modules and the 'clear counters' command to reset the interface counters.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure that it is indeed MALL-ACC-21 in the Malleswaram Node.

- check: Confirm that the vendor-specific CLI dialect is set to Arista.

- check: Ensure that the Access Node (MALL-ACC-21) is reachable and accessible for troubleshooting.

- check: Check the status of the redundant link to ensure that it is operational and can take over in case of a failure.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status: Admin up, Oper down, Description: This command checks the status of the SFP/XFP modules and displays whether they are administratively up or operational down.

Command: show interface counters, Expected Output: Interface counters: Rx bytes: 0, Tx bytes: 0, Description: This command displays the interface counters, including the number of received and transmitted
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bytes.

Command: show log, Expected Output: Log messages: Error: SFP/XFP
module degraded, Description: This command displays the system log
messages, including any error messages related to the SFP/XFP module
degradation.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace module with , Description: This step replaces the degraded SFP/XFP module with a new one, ensuring that the Access Node (MALL-ACC-21) can transmit and receive optical signals correctly.

Step 2: Step: Repair or replace the fiber optic cable, Command: repair fiber optic cable , Description: This step repairs or replaces the fiber optic cable to ensure that the optical signals are transmitted correctly.

Step 3: Step: Adjust the Rx/Tx power settings, Command: adjust power settings , Description: This step adjusts the receive and transmit power settings to ensure that the optical signals are received correctly.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status:
Admin up, Oper up, Description: This command verifies that the SFP/XFP
module is operational and the Access Node (MALL-ACC-21) is
transmitting and receiving optical signals correctly.

Command: show interface counters, Expected Output: Interface counters:
Rx bytes: non-zero, Tx bytes: non-zero, Description: This command
verifies that the interface counters are displaying non-zero values,
indicating that the Access Node (MALL-ACC-21) is transmitting and
receiving data correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore module to , Description: This step restores the original SFP/XFP module to ensure that the Access Node (MALL-ACC-21) is operating with the same configuration as before the fix.

- Step: Revert the fiber optic cable repair, Command: revert fiber optic cable , Description: This step reverts the fiber optic cable repair to ensure that the original cable is used.